

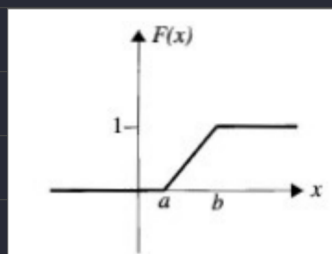
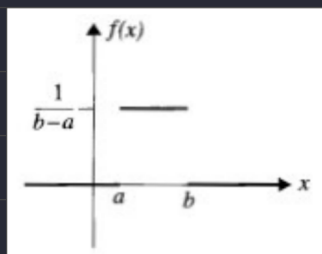
Special Continuous Distributions

7.1 Uniform Random Variable

Definition of Uniform R.V.

在 (a, b) 區間中的機率都是一樣的, 且 uniform = the same

$$f(x) = F'(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases} \quad F(x) = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x \geq b \end{cases}$$



$$E(X) = \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{1}{2} x^2 \right]_a^b = \frac{1}{2(b-a)} (b^2 - a^2) = \frac{a+b}{2}$$

$$E(X^2) = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{1}{3} x^3 \right]_a^b = \frac{1}{3} (a^2 + ab + b^2)$$

$$\therefore \text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{(b-a)^2}{12}$$

7.2 Gaussian (or Normal) R.V.

Definition

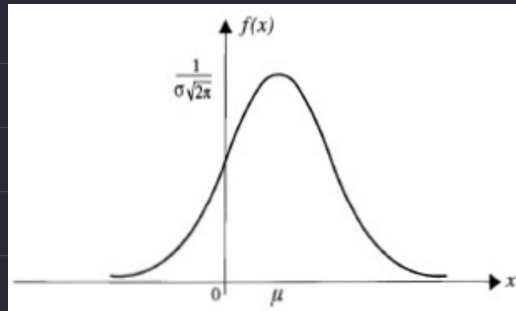
$N(\mu, \sigma^2)$ and the probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Mean value μ

Variance σ^2

Standard Deviation σ



Standard Normal R.V.

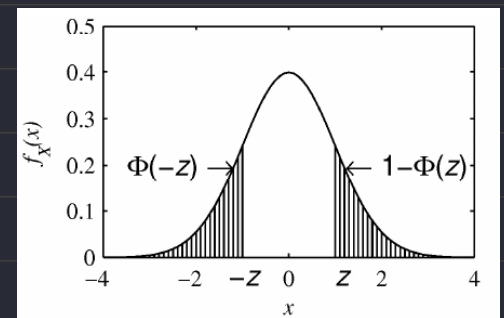
A standard normal random variable Z is a normal random variable with parameters

$N(0, 1)$ $\mu = 0$ $\sigma^2 = 1$

p.d.f: $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$

c.d.f: $P(Z < t) = \Phi(t) = \frac{1}{2\pi} \int_{-\infty}^t e^{-\frac{z^2}{2}} dz$

Note: $\Phi(-z) = 1 - \Phi(z)$



Theorem (Standardized)

For any Gaussian random variable with parameters $N(\mu, \sigma^2)$

the CDF is $F(t) = P(X \leq t) = \Phi\left(\frac{t - \mu}{\sigma}\right)$

and the probability in $[a, b]$ is $P(a < X \leq b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$

Theorem

If X is a Gaussian random variable with parameters $N(\mu, \sigma^2)$

then $E[X] = \mu$ $Var(X) = \sigma^2$ and $Z = \frac{X - \mu}{\sigma}$ is a standard normal R.V

Theorem

If X is a Gaussian random variable with parameters $N(\mu, \sigma^2)$

then $Y = aX + b$, then Y is a Gaussian R.V with parameter $N(a\mu + b, a^2\sigma^2)$

De Moivre-Laplace Theorem

For a Binomial random variable X with parameters $B(n, p)$

$$(1) \quad \lim_{n \rightarrow \infty} P\left(a < \frac{X - np}{\sqrt{np(1-p)}} < b\right) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{t^2}{2}} dt$$

$$\mu = np \quad \sigma = \sqrt{np(1-p)} \quad \rightarrow \text{標準化後收斂到 } N(0, 1)$$

(2)

$$P(k_1 \leq X \leq k_2) \approx \Phi\left(\frac{k_2 + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k_1 - 0.5 - np}{\sqrt{np(1-p)}}\right) \quad k_1, k_2 \text{ are integer}$$

→ 為了讓連續的 Normal R.V. 可以再離散的 Binomial R.V. 可以被計算

所以 $+0.5$ & -0.5 開闢了一條 x 軸空間，避免近似後機率直接 $= 0$

7.3 Exponential Random Variable

Definition

A continuous random variable X is an exponential random variable with parameter $\lambda > 0$, $x > 0$ and:

$$\text{p.d.f. } f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad \text{arrive rate: } \lambda$$

$$\text{c.d.f. } F(t) = \begin{cases} 1 - e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases} \quad \text{arrive time: } t$$

→ This R.V. is used to estimate the time between two event

$$E(X) = \frac{1}{\lambda}$$

$$E(X^2) = \frac{1}{\lambda^2}$$

$$\text{Var}[X] = \frac{1}{\lambda^2}$$

Ex If the duration of a phone call is an exponential r.v. with mean of 10 minutes.

If someone arrives immediately ahead of you at a phone booth, what is the probability that you have to wait

(1) more than 10 minutes ?

(2) between 10 and 20 minutes ?

$$E(X) = \frac{1}{\lambda} = 10 \quad \lambda = \frac{1}{10}$$

$$P(X > 10) = 1 - F(10) = 1 - (1 - e^{-10 \cdot \frac{1}{10}}) = e^{-1}$$

$$P(10 < X < 20) = F(20) - F(10) = 1 - e^{-20 \cdot \frac{1}{10}} - (1 - e^{-10 \cdot \frac{1}{10}}) = e^{-1} - e^{-2}$$

Relation of Exponential and Geometric R.V.

If X is an exponential random variable, then $[X]$, the integer part of X ; i.e, the greatest integer less than or equal to X , is a geometric random variable.

→ 作業有證明

Theorem (Memoryless Property)

Exponential R.V. 會忘記目前的狀態，直接計算條件過後的機率

$$P(X \geq t + s \mid X \geq t) = P(X \geq s) \quad \text{已使已經過了 } t \text{ 年，但是活到 } t+s \text{ 年的機率不受 } t \text{ 影響}$$